

Data collection

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What is Data Collection?

- Research data is any information that has been collected, observed, generated or created to validate original research findings.
- It is the process by which the researcher collects the information needed to answer the research problem
- The task of data collection begins after a research problem has been defined and research design chalked out.

In collecting the data, the researcher must decide

- Which data to collect?
- How to collect the Data?
- Who will collect the Data?
- When to collect the Data?

The selection of a method for collecting information depends upon the

- Resources available
- Credibility
- Analysis and reporting
- Resources, and
- The skills of the evaluator

Methods of Data Collection

- Essentially Two Types:
- Primary Data
 - Primary data are those which are collected for the first time and are original in character.
- Secondary Data
 - Secondary data are those which have already been collected by someone else and which have through some statistical analysis.

Advantages of Primary data

- Primary data is specific to the needs of the researcher at the moment of data collection.
- It is accurate compared to secondary data. The data is not subjected to personal bias and as such the authenticity can be trusted.
- The researcher exhibit ownership of the data collected through primary research. He or she may choose to make it available publicly, patent it, or even sell it.
- Primary data is usually up to date because it collects data in real-time and does not collect data from old sources.
- The researcher has full control over the data collected through primary research. He can decide which design, method, and data analysis techniques to be used.

Disadvantages of Primary data

- Primary data is very expensive compared to secondary data. Therefore, it might be difficult to collect primary data.
- It is time-consuming.
- It may not be feasible to collect primary data in some cases due to its complexity.

Advantages of secondary data

- Secondary data is easily accessible compared to primary data. Secondary data is available on different platforms that can be accessed by the researcher.
- Secondary data is very affordable. It requires little to no cost to acquire them because they are sometimes given out for free.
- The time spent on collecting secondary data is usually very little compared to that of primary data.
- It helps to generate new insights into existing primary data.

Disadvantages of secondary data

- Secondary data may not be authentic and reliable. A researcher may need to further verify the data collected from the available sources.
- Researchers may have to deal with irrelevant data before finally finding the required data.
- Some of the data is exaggerated due to the personal bias of the data source.
- Secondary data sources are sometimes outdated with no new data to replace the old ones.

Sources of secondary data

- Published sources
- Unpublished sources

Collection of Primary Data

- Primary Data may be collected through:
 - Direct personal interview
 - Indirect oral interviews
 - Observation
 - Questionnaires
 - Schedules
 - Information through correspondents

Collection of Primary Data

- Of the above, the important ones are:
 - Observation Method
 - Interview Method
 - Questionnaires
 - Schedules

Observation Method

- Observation method is a method under which data from the field is collected with the help of observation by the observer or by personally going to the field.
- In the words of P.V Young “Observation may be defined as systematic viewing, coupled with consideration of seen phenomenon.”

Steps For An Effective Observation

Steps

Determine what needs to be observed



Select participants
(Random/non-random)



Conduct the observation (venue, duration, recording materials, take photographs)



Compile data collected



Analyze and interpret data collected

Classification-1

- Structured Observation
 - When the observation is characterized by a careful definition of the units to be observed, the style of recording the observed information, standardized conditions of observation and the selection of related data of observation.
- Unstructured Observation
 - When it takes place without the above characteristics.

Classification-2

- Participant Observation
 - When the observer is member of the group which s/he is observing then it is Participant Observation.
- Non-Participant Observation
 - When observer is observing people without giving any information to them then it is Non-Participant Observation.

Classification-3

- Uncontrolled Observation
 - When the observation takes place in natural condition i.e., uncontrolled observation.
 - It is done to get spontaneous picture of life and persons.
- Controlled Observation
 - When observation takes place according to pre arranged plans, with experimental procedure then it is controlled observation generally done in laboratory under controlled condition.

Interview method

- Interview is the verbal conversation between two people with the objective of collecting relevant information for the purpose of research.
- According to McNamara-
 - Interviews are particularly useful for getting the story behind a participant's experiences.
 - The interviewer can pursue in-depth information around the topic.

TYPES OF INTERVIEW

- Personal Interview
- Telephone Interview
- Focus Group Interview
- Depth Interview

Personal Interview

- Personal Interview communication is between a face the to face two way interviewer and the respondents.
- Generally the personal interview is carried out in a planned manner and is referred to as 'structured interview'.

Telephone Interview

- Telephone interview the information is collected from the respondent by asking him questions on the phone is called as telephone interview.
- The combination of telephone and computer has made this method even more popular.
- It has certain advantages and disadvantages.

Focus Group Interview

- Focus group interview is an unstructured interview which involves a moderator leading a discussion between a small group of respondents on a specific topic.

- Focus group interview results in advantages summed up as 10s. they as follow.

- Synergism – working together of two things
- Snowballing – Collects information as it transfers from one person to another
- Stimulation – inspires the data collection
- Security- Security of information and its authenticity.
- Spontaneity - Information collected would be natural
- Serendipity – chance of getting relevant information to the study.
- Specialization- Specialized information
- Scientific scrutiny- Selected informants so highly scientific
- Structure- Question are already structured
- Speed- Speedy in data collection

Depth Interview

- Depth interview is nondirective in nature where the respondent is given freedom to answer within the boundaries of the topic of interest.
- During in-depth interviews, researchers and participants have the freedom to explore additional points and change the direction of the process when necessary. It is an independent research method that can adopt multiple strategies according to the needs of the research.

Steps to conduct in-depth interviews

- Obtain the necessary information about the respondents and the context in which they operate.
- Make a script or a list of topics you want to cover. This will make it easy to add secondary questions.
- Schedule an interview at a time and date of the respondent's choice.
- Ask questions confidently and let the interviewees feel comfortable, so that they too are confident and can answer difficult questions with ease.
- Set a maximum duration such that it doesn't feel exhaustive.
- Observe and make notes on the interviewee's body expressions and gestures.
- It is important to maintain ethics throughout the process.
- Transcribe the recordings and verify them with the interviewee.

Advantages of Interview Method

- Opportunity for Feedback – Interviewer can provide direct feedback to the respondent, give clarifications .
- Probing Complex Answers – Interviewers can investigate if the respondent's answer is too brief or unclear. This gives interviewers some flexibility in dealing with unstructured questions is especially suited for handling complex questions.
- Length of Interview – If the questionnaire is very lengthy, the personal interview is the best technique for getting respondents to cooperate, without overtaxing their patience

- Complete answer of the Questions – Respondent will answer all questions asked, unlike in telephone interview where the respondent may hang up or in mail questionnaire where some questions may go unanswered.
- High indexing Participation – Interviewing respondents personally can increase the likelihood of their participation, many people participate directly.

Disadvantages of Interview Method

- Cost – Personal interviews are usually more expensive than mail, telephone and internet surveys.
 - Factors influencing the cost of the interview include the respondents' geographic proximity, the length and complexity of the questions, and the number of non-respondents
- Lack of Anonymity – Respondents are not anonymous in a personal (face-to-face) interview and may be unwilling to disclose certain information to the interviewer.
- Necessity for Callbacks – When a person selected for interview cannot be reached the first time, a callback has to be scheduled which result in extra cost and time spent.

- Variance Effects – It has been shown that the demographic characteristics of the interviewer can influence the answers of the respondents. In one study, male interviewers had a much larger variance of answers than female interviewers in a sample of most female individuals
- Dishonesty – Interviewers may cheat to make their life easier and save time and effort
- Cultural Considerations – Cultural aspects may influence peoples' willingness to participate in an interview (e.g. some cultures still discourage females from being questioned by male interviewers)

Questionnaire

- The questionnaire was invented by Sir Francis Galton, a British anthropologist, explorer and statistician, invented questionnaires in the late 1800s.
- A tool for collecting information to describe, compare, or explain an event or situation, as well as, knowledge, attitudes, behaviors, and/or sociodemographic characteristics on a particular target group.

- The method of data collection is one of the important steps in the research process.
- Questionnaire is one such tool used to collect information which can cover almost all practical issues
- Questionnaires are extensively used in surveys and forms the backbone of the survey procedure

Questionnaire General Format

- Self-administered (by mail or personal contact)
- In person (face-to-face)
- Telephone interviews

Questions Context

- Identify the questionnaire's specific purpose(s)
- Clarify the terms used to state the questionnaire's purpose(s)
- Be sure to have the specific objectives of the questionnaire
- Know the respondents

Questions Format

- Open questions - more information but difficult to codify, enter, and analyze
- Closed questions - less information but easy to codify, enter, and analyze

How To Obtain Valid Information

- Ask purposeful questions
- Ask concrete questions
- Use time periods based on importance of the questions
- Use conventional language
- Use complete sentences

- Avoid abbreviations
- Review questions with experts and potential respondents
- Use shorter questions
- Avoid two-edged questions
- Avoid negative questions
- Adopt questions used successfully in other questionnaires

ADVANTAGES OF QUESTIONNAIRE

- Low cost even when the universe is large and is widely spread geographically.
- It is free from the bias of the interviewer; answers are in respondents own words.
- Respondents have adequate time to give well thought out answers
- Respondents who are not easily approachable can also be reached conveniently.
- Large samples can be made use of and thus the results can be made more dependable and reliable.

DISADVANTAGES OF QUESTIONNAIRE

- Low rate of return of the duly filled in questionnaires
- It can be used only when respondents are educated and co- operating.
- The control over the questionnaire may be lost once it is sent
- It is difficult to know whether willing respondents are truly representative
- There is also the possibility of ambiguous replies or omission of replies altogether to certain questions
- This method is likely to be the slowest of all
- Respondents may misinterpret a question, thereby limiting the validity of the results

HOW TO OVERCOME

- Pilot testing the questionnaire with individuals who are similar to the sample population who have been chosen to participate in the study can minimize the disadvantage.
- These individuals can alert the investigator to any unclear items.
- After reviewing the results from the pilot study the researcher can revise the questionnaire accordingly before distribution to the population sample.
- Clear instructions should be provided for respondents explaining how to complete the questionnaire and requesting that all items be answered completely.

FORMS OF QUESTIONNAIRE

- The general form of questionnaire can be three types.
 - Structured
 - Semi-structured
 - Unstructured

STRUCTURED QUESTIONNAIRE

- Structured questionnaires are those questionnaires in which there are definite, concrete and pre-determined questions.
- The questions are presented with exactly the same wording and in the same order to all respondents.
- Resort is taken to this sort of standardization to ensure that all respondents reply to the same set of questions
- Structured questionnaires have fixed alternative questions in which responses of the informants are limited to the stated alternatives

- A highly structured questionnaire is one in which all questions and answers are specified and comments in the respondents own words are held to minimum.
- Structured questionnaires are simple to administer and relatively inexpensive to analyse.
- They are used in large interview programmes (anything over 30 interviews and more likely over 200 interviews in number) and may be carried out over the tele- phone, face-to-face or self completion depending on the respondent type, the content of questionnaire and the budget.

SEMI-STRUCTURED QUESTIONNAIRES

- Semi-structured questionnaires comprise a mixture of closed and open questions.
- They are commonly used in research where there is a need to accommodate a large range of different responses from companies.
- The use of semi-structured questionnaires enables a mix of qualitative and quantitative information to be gathered.
- They can be administered over the telephone or face-to-face.

UNSTRUCTURED QUESTIONNAIRES

- Unstructured questionnaires are made up of questions that elicit free responses.
- These are guided conversations rather than structured interviews and would often be referred to as a “topic guide”.
- The topic guide is made up of a list of questions with an apparent order but is not so rigid that the interviewer has to slavishly follow it in every detail.

- There are no hard-and-fast rules about how to design a questionnaire, but there are a number of points that can be borne in mind:
 - A well-designed questionnaire should meet the research objectives.
 - It should obtain the most complete and accurate information possible.
 - A well-designed questionnaire should make it easy for respondents to give the necessary information and for the interviewer to record the answer, and it should be arranged so that sound analysis and interpretation are possible.
 - It would keep the interview brief and to the point and be so arranged that the respondent(s) remain interested throughout the interview.

Steps involved in the development of a questionnaire

- Decide the information required.
- Define the target respondents.
- Choose the method(s) of reaching your target respondents.
- Decide on question content.
- Develop the question wording.
- Put questions into a meaningful order and format.
- Check the length of the questionnaire.
- Pre-test the questionnaire.
- Develop the final survey form.

Qualitative research

- Qualitative research involves collecting and analyzing non-numerical data (e.g., text, video, or audio) to understand concepts, opinions, or experiences. It can be used to gather in-depth insights into a problem or generate new ideas for research. Qualitative search is subjective in nature.

Qualitative research methods

- **Observations:**
 - recording what you have seen, heard, or encountered in detailed field notes.
- **Interviews:**
 - personally asking people questions in one-on-one conversations.
- **Focus groups:**
 - asking questions and generating discussion among a group of people.
Generally between 8-12 people.
- **Survey:**
 - distributing questions with open-ended questions.
- **Secondary research:**
 - collecting existing data in the form of texts, images, audio or video recordings, etc.

CONSIDERATIONS FOR COLLECTING DATA

- Accuracy
 - The data collection technique that will provide the most accurate results is desired when selecting a data collection method.
- Validity and Reliability
 - Reliability and validity are both about how well a method measures something: Reliability refers to the consistency of a measure (whether the results can be reproduced under the same conditions). Validity refers to the accuracy of a measure (whether the results really do represent what they are supposed to measure)
- Time and Cost
 - When selecting data collection methods, several issues should be considered with regard to time and cost. The time required to complete the instrument is one consideration. Also, consider the time required for managers to complete the instrument if they are involved, or the time in assisting participants through the data collection process.
- Utility
 - A final consideration when selecting a data collection method is utility. How useful will the data be, given the type of data collected through the process?

Steps of Data Processing and Presentation

- **Data processing** is concerned with editing, coding, classifying, tabulating and charting and diagramming research data. The essence of **data processing in research** is data reduction.
- Data reduction involves winnowing out the irrelevant from the relevant data and establishing order from chaos and giving shape to a mass of data.
- **Data processing in research** consists of five important steps. They are: Editing of Data, Coding of data, Classification of data, Tabulation of data, Diagramming the data.

Editing of Data

- Editing of Data Editing is the first step in data processing. Editing is the process of examining the data collected in questionnaires/schedules to detect errors and omissions and to see that they are corrected and the schedules are ready for tabulation. When the whole data collection is over a final and a thorough check up is made. The editor is responsible for seeing that the data are;
 - Accurate as possible
 - Consistent with other facts
 - Uniformly entered
 - As complete as possible
 - Acceptable for tabulation and arranged to facilitate coding tabulation.

Coding of data

- Coding is the process/operation by which data/responses are organized into classes/categories and numerals or other symbols are given to each item according to the class in which it falls. In other words, coding involves two important operations;
 - Deciding the categories to be used and
 - Allocating individual answers to them.

Classification of Data

- Classification or categorization is the process of grouping the statistical data under various understandable homogeneous groups for the purpose of convenient interpretation.
- A uniformity of attributes is the basic criterion for classification; and the grouping of data is made according to similarity.
- Classification becomes necessary when there is a diversity in the data collected for meaningless for meaningful presentation and analysis.
- However, it is meaningless in respect of homogeneous data. A good classification should have the characteristics of clarity, homogeneity, equality of scale, purposefulness and accuracy.

Tabulation of Data

- Tabulation is the process of summarizing raw data and displaying it in compact form for further analysis. Therefore, preparing tables is a very important step. Tabulation may be by hand, mechanical, or electronic. The choice is made largely on the basis of the size and type of study, alternative costs, time pressures, and the availability of computers, and computer programmes. If the number of questionnaire is small, and their length short, hand tabulation is quite satisfactory.
- Table may be divided into: (i) Frequency tables, (ii) Response tables, (iii) Contingency tables, (iv) Univariate tables, (v) Bivariate tables, (vi) Statistical table and (vii) Time series tables.

Data Diagrams

- Diagrams are charts and graphs used to present data. These facilitate getting the attention of the reader more. These help presenting data more effectively. The data diagrams classified into:
- Charts: A chart is a diagrammatic form of data presentation. Bar charts, rectangles, squares and circles can be used to present data.
- Graphs: The method of presenting numerical data in visual form is called graph, A graph gives relationship between two variables by means of either a curve or a straight line. Graphs may be divided into two categories. (1) Graphs of Time Series and (2) Graphs of Frequency Distribution. In graphs of time series one of the factors is time and other or others is / are the study factors. Graphs on frequency show the distribution of by income, age, etc. of executives and so on.